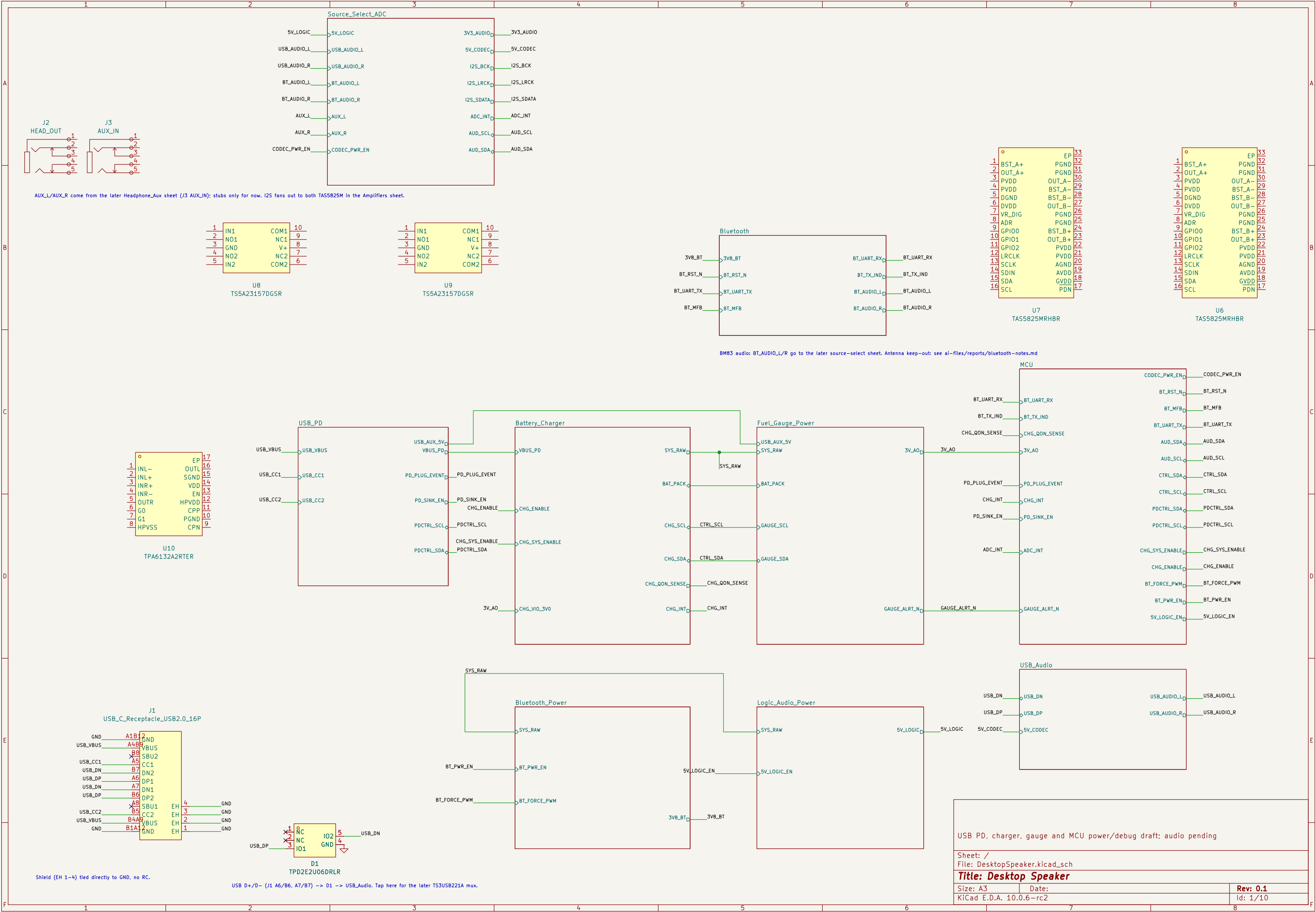
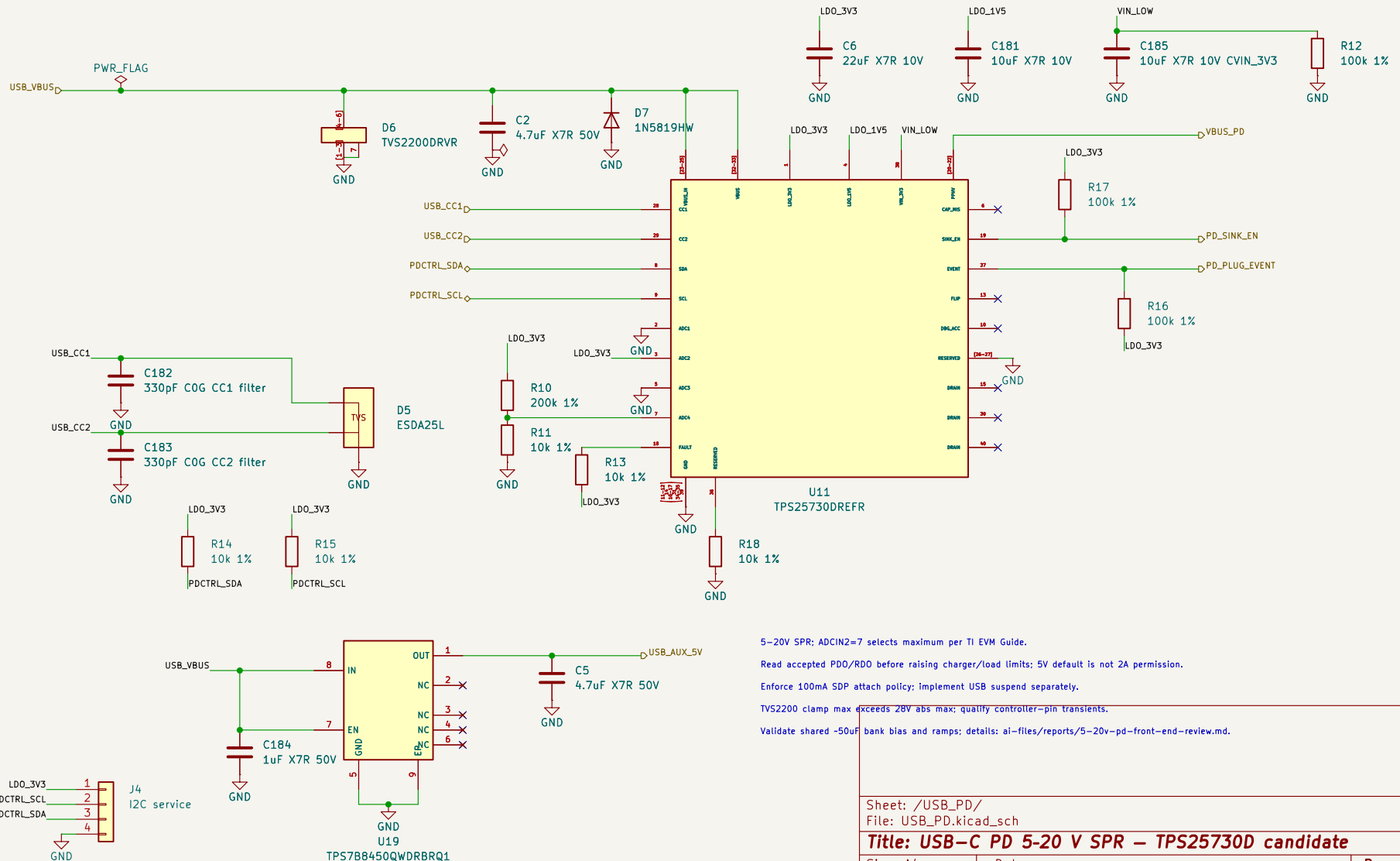






5-20V SPR; ADCIN2=7 selects maximum per TI EVM Guide.

Read accepted PDO/RDO before raising charger/load limits; 5V default is not 2A permission.

Enforce 100mA SDP attach policy; implement USB suspend separately.

TVS2200 clamp max exceeds 28V abs max; qualify controller-pin transients.

Validate shared -50uF bank bias and ramps; details: ai-files/reports/5-20v-pd-front-end-review.md.

Sheet: /USB_PD/

File: USB_PD.kicad_sch

Title: USB-C PD 5-20 V SPR - TPS25730D candidate

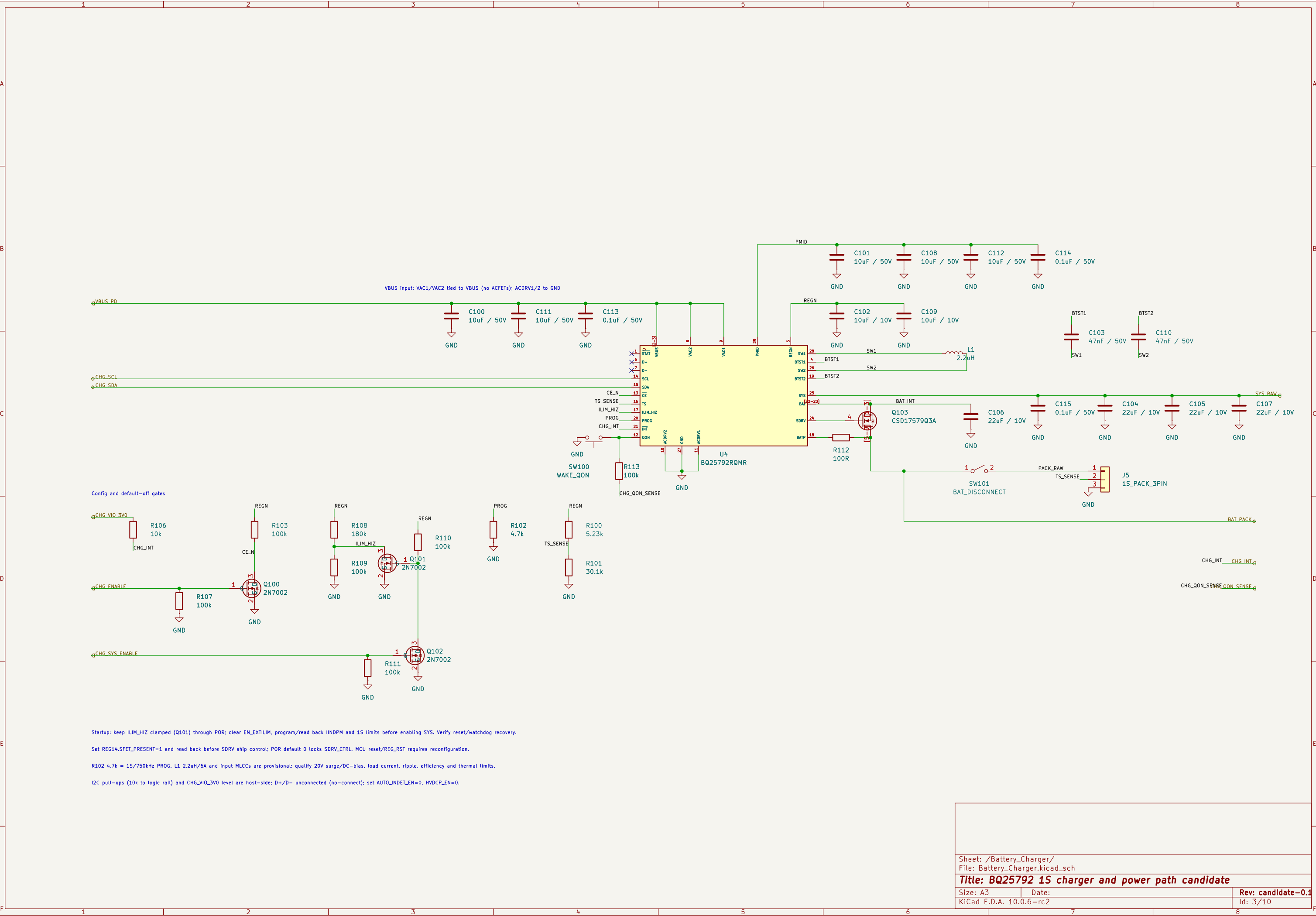
Size: A4

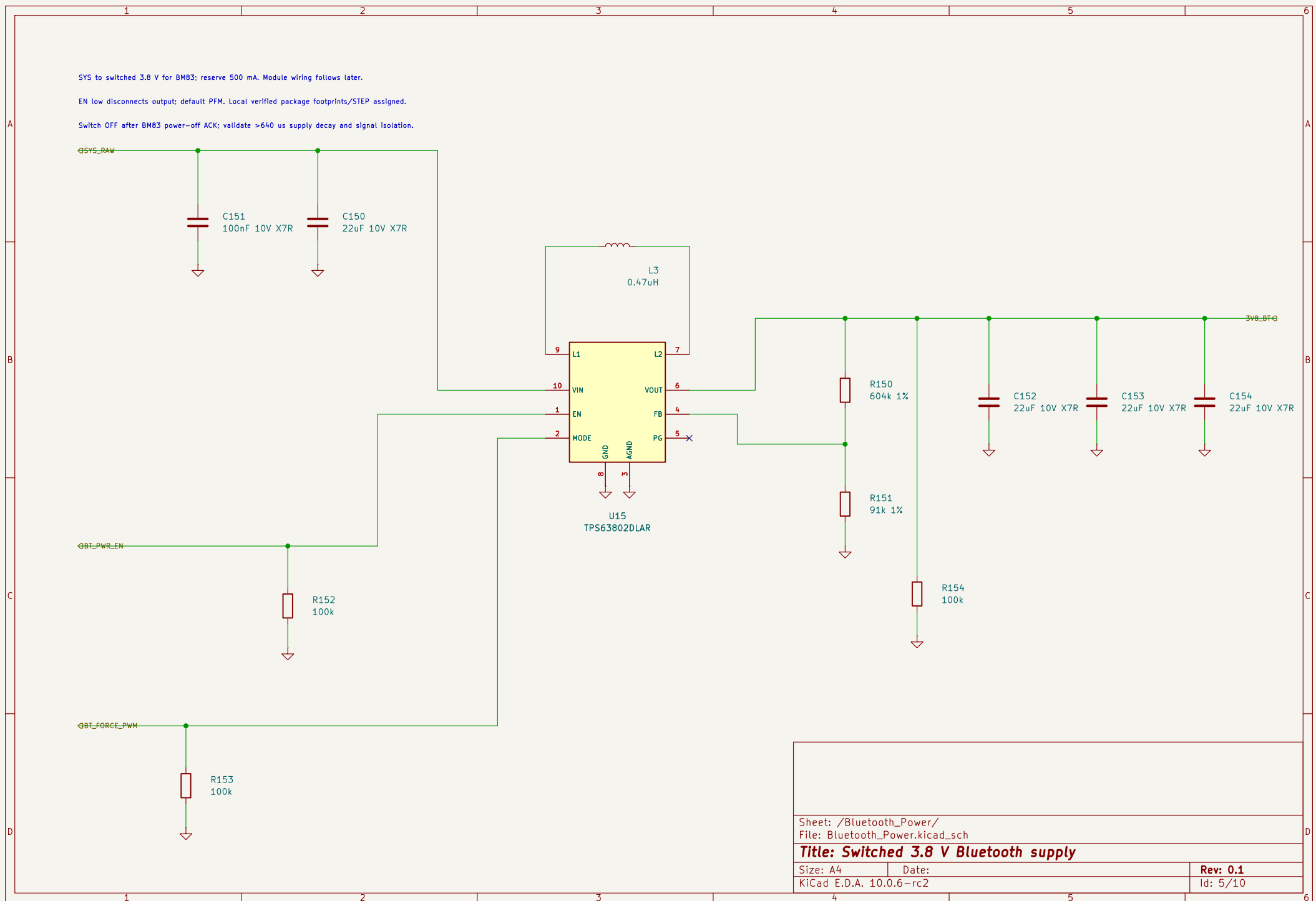
Date:

KiCad E.D.A. 10.0.6-rc2

Rev: prototype review

Id: 2/10

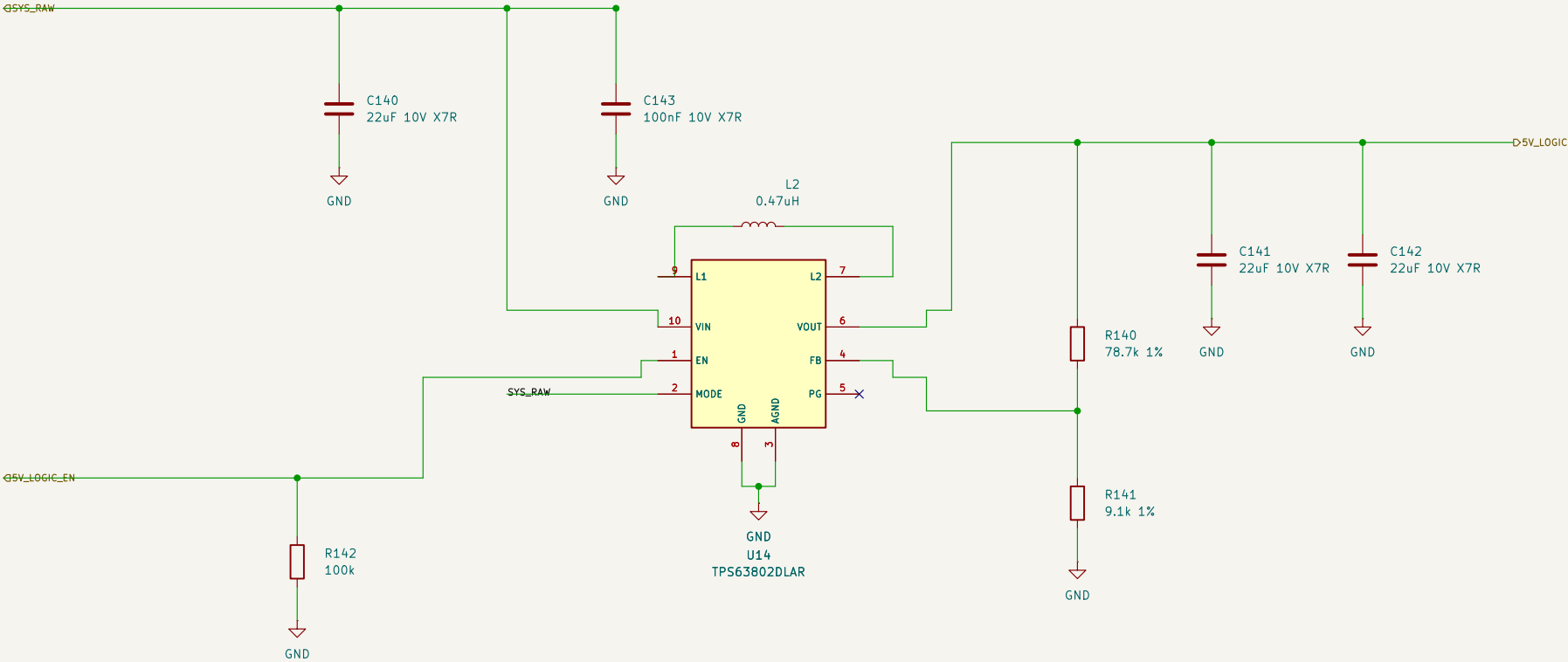




Protected 1S SYS to switched 5 V; reserve 500 mA total. Codec and other 5 V loads connect later.

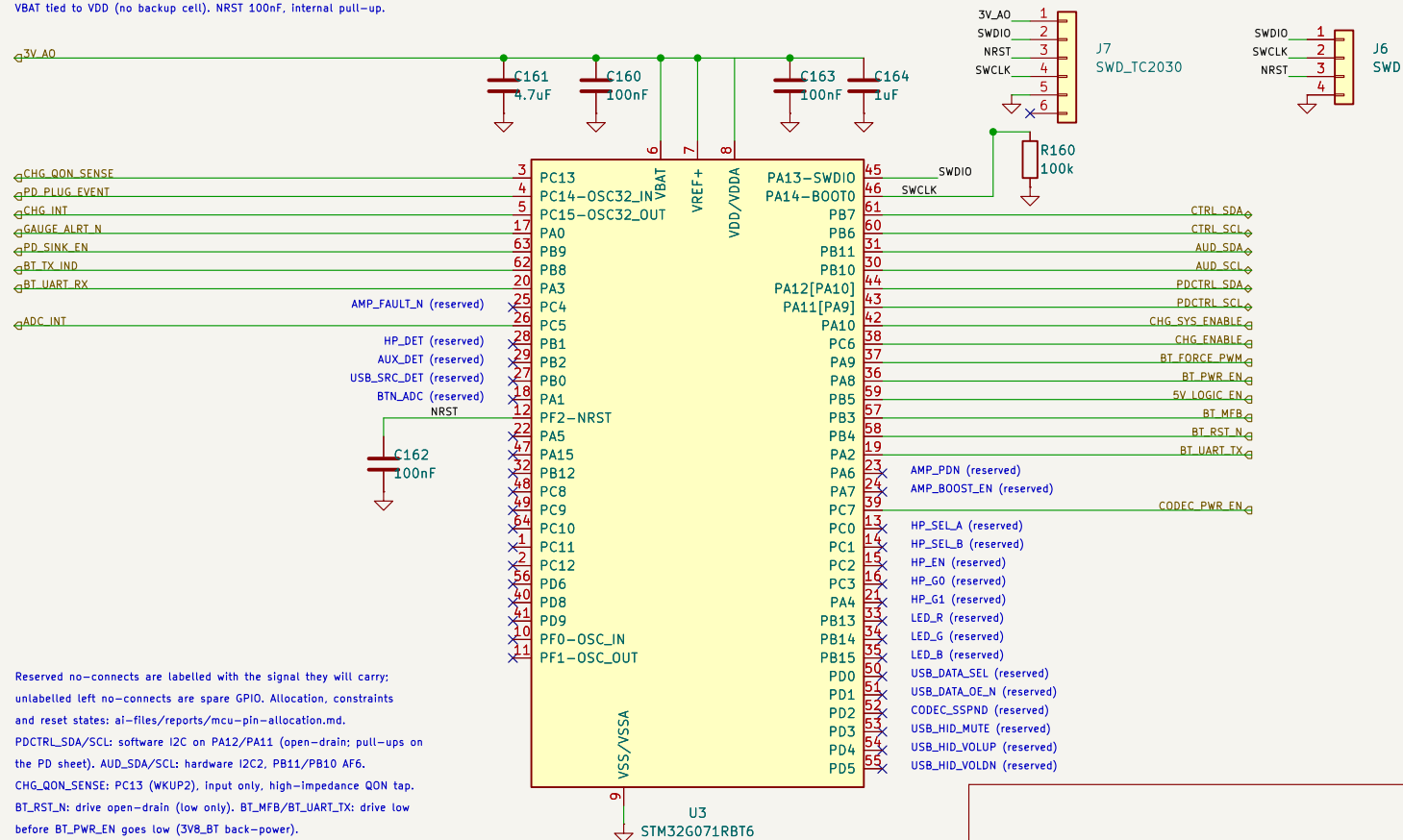
EN low = true input/output disconnect; MODE tied to SYS_RAW forces PWM after soft-start.

Steady PWM divider DC bound is within PCM2902C limits; soft-start uses PFM. Ripple and startup transient require qualification.



3V_AO supply: VDD/VDDA 100nF + 4.7uF (DS12232 fig. 13). VREF+ 100nF + 1uF (VREFBUF off, VREF+ = VDDA).
VBAT tied to VDD (no backup cell), NRST 100nF, internal pull-up.

SWD: J7 Tag-Connect TC2030-IDC (with legs: 6 SWO unused) and J6 4-pin 2.54 mm header share SWDIO, SWCLK, NRST
R160 holds PA14/BOOT0 low (nBOOT_SEL=1 ignores the pin)



STM32G071RBT6 LQFP64: supply, reset, SWD, power-block links

Sheet: /MCU/
File: MCU.kicad_sch

Title: MCU and programming

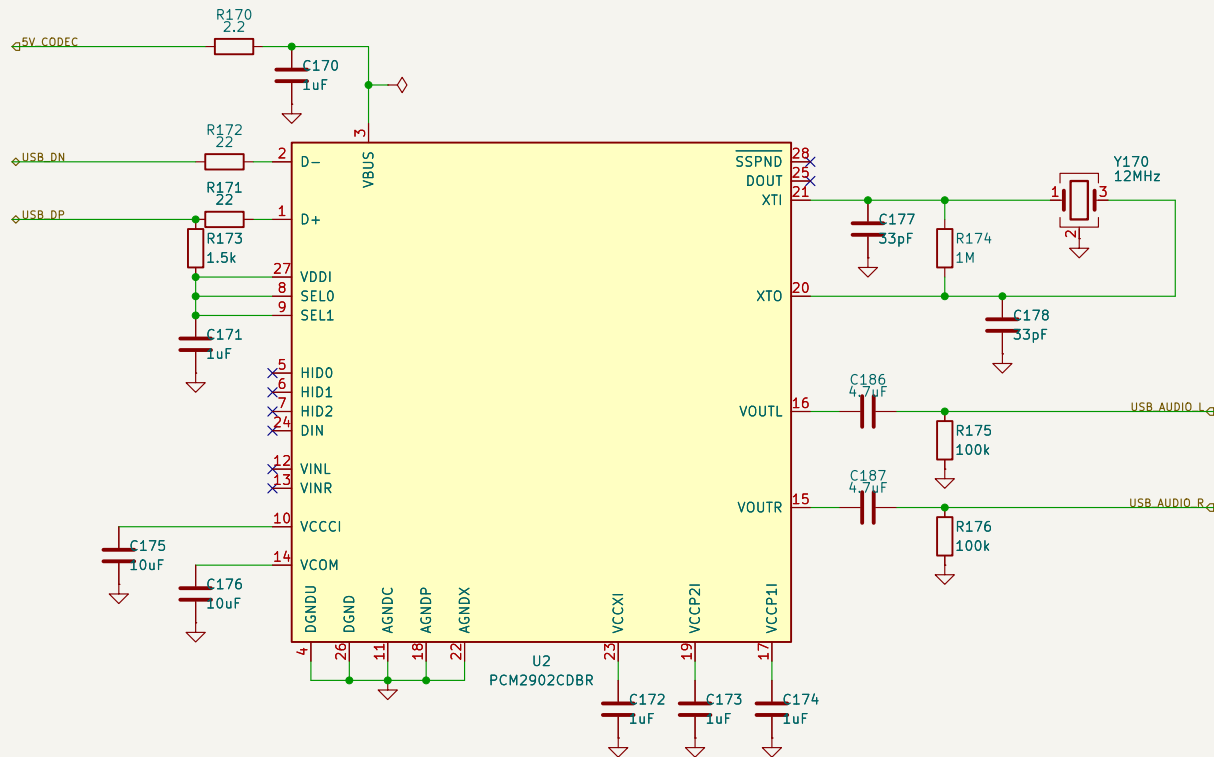
Size: A4
KiCad E.D.A. 10.0.6-rc2

Date:

Rev: 0.1

Id: 7/10

Supply: 5V_CODEC (U23 load switch from 5V_LOGIC, 4.60–5.06 V worst case) feeds VBUS through 2.2R/1uF. DS range 4.35–5.25 V, 56 mA typ / 67 mA max, 250 uA suspend.
 Codec is powered only when CODEC_PWR_EN (U23) and 5V_LOGIC are on: D+ pull-up (R173 to VDDI) = USB attach. Do not enable CODEC_PWR_EN on DCP/unknown sources.
 SEL0/SEL1 = high (VDDI). HID0–2, DIN, VINL/R unused. SSPND (suspend flag, active low) reserved for a later MCU input.
 USB_DP/USB_DN come from J1 via D1 (root). Later source-detect mux goes between those root nets and these ports.



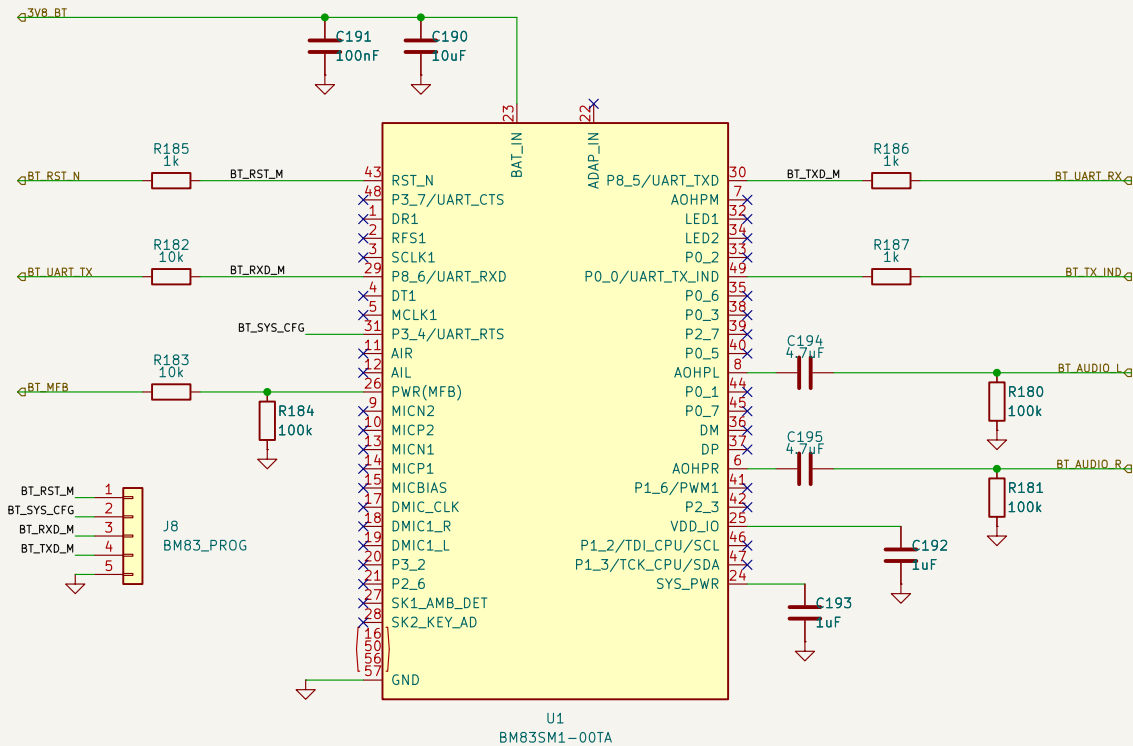
Analogue out: 0.6 VCCCI Vpp (about 2.0 Vpp, 0.7 Vrms) centred on 1.65 V; AC-coupled (HPF -0.4 Hz at 100k), >=10k load.

PCM2902C bus-powered codec from 5V_LOGIC; USB D+/D- path; analogue out to USB_AUDIO_L/R

Sheet: /USB_Audio/
 File: USB_Audio.kicad_sch

Title: USB audio

Size: A4	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2	Id: 8/10	



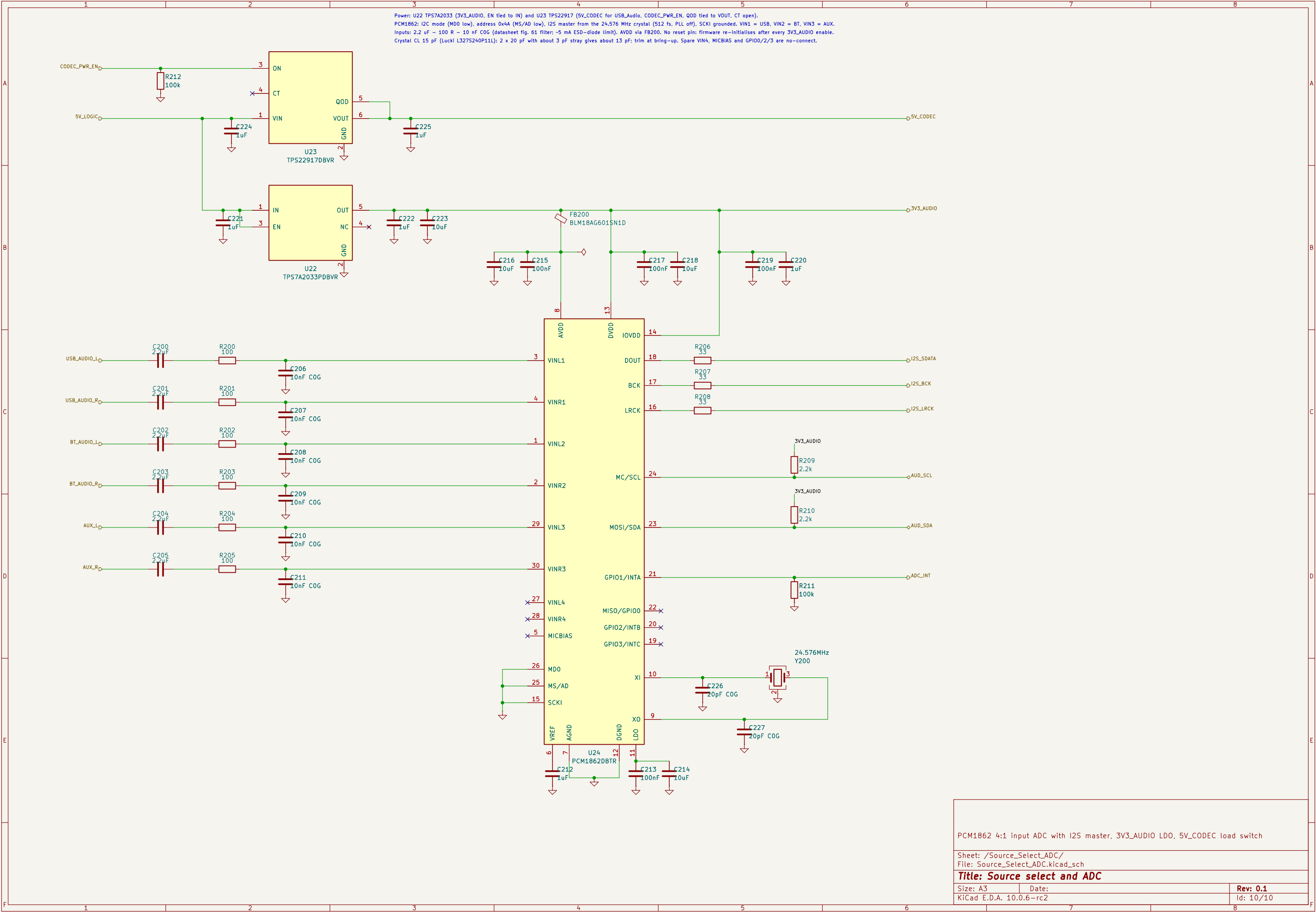
BM83SM1-00TA (DS70005402D). BAT_IN 3.2-4.2 V from 3V8_BT (U15); SYS_PWR/VDD_IO are outputs (decoupling only, do not load or feed). ADAP_IN unused (Internal charger/USB DFU not used).
Host mode UART from the MCU: PB_6 RXD / PB_5 TXD, P0_0 = UART_TX_IND (wakes the MCU). MFB (PWR) = power-on/wake key. RST_N = reset (active low, internal pull-up). Config Tool must set Host mode, P0_0 as TX_IND.
Back-power: MCU lines pass 10k/1k series resistors; firmware must drive PA2/PB3 low (PB4 open-drain/hi-Z) and disable BT_PWR_EN only after the module power-off ACK. Module outputs are 3.0-3.6 V into 5 V tolerant MCU pins.
Audio: single-ended DAC out (AOHPL/AOHPM) AC-coupled, 4.7 uF + 100k bleed, 0.74 Vrms full scale; AOHPM (capless sense) unused. Ports feed the later source-select sheet.
Antenna: integrated PCB antenna, no external RF parts. Keep the antenna end of the module off the board (no copper/components/keep-out) and metal 15 mm clear; see ai-files/reports/bluetooth-notes.md.
J8 programming/test (isUpdate/Config Tool): 1 RST_N, 2 P3_4 SYS_CFG (low at reset = test mode), 3 UART_RXD (adapter TX), 4 UART_TXD (adapter RX), 5 GND. 3V8_BT must be enabled first.

BM83 module: host UART, MFB/RST_N, analogue out

Sheet: /Bluetooth/
File: Bluetooth.kicad_sch

Title: Bluetooth audio

Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 9/10



PCM1862 4:1 input ADC with I2S master, 3V3_AUDIO LDO, 5V_CODEC load switch